Team XX – Name 1 (Roll), Name 2 (Roll), Name 3 (Roll), Name 4 (Roll) `https://github.com/<team>/ffs-diverse`

1 Summary of the Assigned Paper

Video-language models are costly because every frame goes through a large model, so systems sample a few frames uniformly and can miss the moment that matters. FFS [1] puts a small selector over cheap CLIP features in front of the large model. It picks question-relevant frames, and learnable `[CTRL]` tokens inside a differentiable top-k let it decide *how many* ($k{-}c$ to k) each question needs. It learns from the answer loss alone, with no frame labels, and keeps dense-frame accuracy at a fraction of the compute.

2 Proposed Novelty: Redundancy-Aware Selection (RAS)

FFS scores frames independently, so a set's value is additive, $\sum_{i\in A} l_i$. Neighbouring frames are near-duplicates with near-equal scores, so top-k spends its tiny budget on repeats, and `[CTRL]` judges a frame by relevance, not by what it *adds*. We replace top-k by greedy selection with a learned redundancy penalty, using a video-adaptive similarity of frozen CLIP features f_i:

$$S_{ij} = \tfrac{[\cos(f_i,f_j)-\mu]_+}{1-\mu}, \qquad i_t = \arg\max_{i\notin A_{t-1}} \big(\underbrace{l_i+\gamma_i}_{\text{FFS}} - \beta \textstyle\sum_{j\in A_{t-1}} S_{ij}\big),\ t = 1..k, \tag{1}$$

where μ is the video's mean off-diagonal cosine, $S{=}0$ for `[CTRL]`, $A_0 = \varnothing$, and γ_i is Gumbel noise (training only); picked `[CTRL]`s are dropped as in FFS. Gradients use the relaxed subset sampler [2] with the same penalty, $w \leftarrow w + \log(1-p) - \frac{\beta}{\tau} Sp$, $p = \mathrm{softmax}(w)$, plus a straight-through estimator; $\beta = \mathrm{softplus}(\rho)$ is learned. The *only* difference from our FFS re-implementation is β ($\beta = 0$ gives FFS exactly). Both share one recipe: count supervision from fixed-k policies (a `[CTRL]` hinge to $k^\star$, the fewest frames whose answer loss is within ϵ of the best) and task gradients restricted to frames.

3 Theoretical Justification

Diminishing returns. Eq. (1) is the greedy step for $F(A) = \sum_{i\in A} l_i - \beta \sum_{\{i,j\}\subseteq A} S_{ij}$, with marginal gain $\Delta(i \mid A) = l_i - \beta \sum_{j\in A} S_{ij}$. Because $S \geq 0$, $A \subseteq B \Rightarrow \Delta(i|A) \geq \Delta(i|B)$, so F is submodular. F also has the shape of a DPP objective [3]: with quality $q_i = e^{l_i}$ and unit-diagonal kernel K, $\log\det L_A = 2\sum_A l_i + \log\det K_A \approx 2\sum_A l_i - \sum_{\{i,j\}\subseteq A} K_{ij}^2$ (second order). That is relevance minus pairwise redundancy, and RAS is its greedy (MMR-style) analogue with S_{ij} in place of K_{ij}^2. So it picks frames that are relevant *and* diverse. **Adaptive budget.** A frame is admitted at step t only if $\max_i \Delta(i|A_{t-1})$ beats the next unused `[CTRL]` score $c_{(r)}$. FFS compares raw relevance l_i with $c_{(r)}$, so it counts *relevant* frames. RAS counts *relevant and new* frames. For the same scores, $s_{\mathrm{RAS}} \leq s_{\mathrm{FFS}}$: each of the m RAS frames had $l_i \geq \Delta \geq c_{(k-m+1)}$, so FFS admits at least m. They are equal e.g. when the picks are mutually dissimilar. The saving grows with video redundancy, where uniform sampling and FFS waste compute. **Safe extension.** $\beta{=}0$ is FFS and β is learned, so training can switch the penalty off if useless.

4 Comparative Results

NExT-QA val (4,996 Qs), EVA02-CLIP-L/14 features, 32-frame pool, one shared frozen downstream model, 3 seeds.

Method	Frames	Acc	C	T	D	Red.↓	Cost
Uniform@4	4.00	–	–	–	–	–	0.125
Uniform@32	32.00	–	–	–	–	–	1.000
Fixed top-4	4.00	–	–	–	–	–	0.125
FFS (re-impl.)	–	–	–	–	–	–	–
FFS+RAS (ours)	–	–	–	–	–	–	–

Tab. 1: NExT-QA val accuracy (%) by question type (C/T/D), avg. frames given to the downstream model, redundancy of picks, and downstream cost relative to 32 frames. Learned β = TODO.

TODO: 2–3 sentences of interpretation (template below). Wider coverage should help multi-event temporal/causal questions; fine-motion questions that need consecutive frames may lose.

References

[1] S. Buch, A. Nagrani, A. Arnab, C. Schmid. Flexible Frame Selection for Efficient Video Reasoning. CVPR 2025.

[2] S. M. Xie, S. Ermon. Reparameterizable Subset Sampling via Continuous Relaxations. IJCAI 2019.

[3] A. Kulesza, B. Taskar. Determinantal Point Processes for Machine Learning. FnT ML 2012.